MAST20026 REAL ANALYSIS 7B

One-Sided Limits, Limits to Infinity and Boundedness - Solutions

A Note on the Solutions: This solution sheet is intended for you verify correctness of your work during the tutorial. The solutions here do not necessarily meet the standards for exposition expected on assignment submissions.

1. **One-Sided Limits**

   Let
   $$f(x) = \begin{cases} 2x & x > 1 \\ -2x & x < 1 \\ 0 & x = 1 \end{cases}$$

   (a) Using an $\epsilon - \delta$ argument, prove $\lim_{x \to 1^-} f(x) = -2$.

   Solution: Let $\delta = \epsilon/2$.

   (b) Using an $\epsilon - \delta$ argument, prove $\lim_{x \to 1^+} f(x) = 2$.

   Solution: Let $\delta = \epsilon/2$.

   (c) By citing a theorem from the lecture notes, prove $f(x)$ diverges at $x = 1$.

   Solution: The result follows directly from Corollary 5.8.

2. **Sequences and Functions Part I**

   Let $f(x) = \frac{4x}{x+3}$. For every $n \in \mathbb{N}^+$, let $f_n = f(n)$.

   (a) Using the definition of convergence as x goes to ∞, prove
   $$\lim_{x \to \infty} f(x) = 4$$

   Solution: We seek $M \in \mathbb{R}$ so that $|f(x) - 4| < \epsilon$ whenever $x > M$. Choose $M = \frac{12}{\epsilon}$.

   (b) In the previous part, you have shown that for every $\epsilon > 0$, there exists $M \in \mathbb{R}$ so that $|f(x) - 4| < \epsilon$ whenever $x > M$. Using this fact, prove $\lim_{x \to \infty} f_n = 4$

   Solution: Choose $M = \lceil \frac{12}{\epsilon} \rceil$ in the definition of convergence for sequences.

3. **Sequences and Functions Part II**

   Let $f : \mathbb{R} \to \mathbb{R}$ be a function and define a sequence (f_n) so that $f_n = f(n)$ for all $n \in \mathbb{N}^+$.

   (a) Let $L \in \mathbb{R}$. Using the definition of convergence for sequences and functions prove that if $\lim_{x \to \infty} f(x) = L$, then $\lim_{n \to \infty} f_n = L$.

   Solution: Let $\epsilon > 0$. Since $\lim_{x \to \infty} f(x) = L$, there exists $M^\star \in \mathbb{R}$ so that $|f(x) - L| < \epsilon$ whenever $x > M^\star$. Let $M = \lceil M^\star \rceil$. Then for all integers $n > M$, $|f(n) - L| < \epsilon$ (since $n > M \geq M^\star$). Therefore, $\lim_{n \to \infty} f_n = L$.

   (b) Notice that the convergence of (f_n) depends only on the behaviour of $f(x)$ for $x \in \mathbb{N}^+$. Give an example of a function $f : \mathbb{R} \to \mathbb{R}$ so that f_n converges as x goes to ∞, but $f(x)$ diverges as x goes to ∞.

   Solution: Consider $f(x) = \sin(2\pi x)$, for example.

4. **Limits and Boundedness**

   In our study of sequences, we proved that every convergent sequence is bounded. Here we consider an analogue of this result for functions approaching a limit point. We prove that if $f(x)$ converges as x approaches a, then $f(x)$ is *bounded* when x is *sufficiently close* to a.

   (a) Let E be a set, let a be a limit point of E and let $\delta > 0$. Let $N_0(a, \delta)$ denote the following set
   $$N_0(a, \delta) = \{x \in \mathbb{R} \mid 0 < |x - a| < \delta\}$$

   Using the definition of limit point, explain how you know $N_0(a, \delta)$ is always non-empty.

   Solution: The definition of limit point tells us that for every δ there exists $x \in E$ so that x is a solution for $0 < |x - a| < \delta$. Therefore $N_0(a, \delta)$ is non-empty.

(b) Let $L \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ and $a \in \mathbb{R}$. Prove that if $\lim_{x \to a} f(x) = L$, then there exists δ so that the set

$$\{f(x) \mid x \in N_0(a, \delta)\}$$

is bounded above and below in $\mathbb{R}$. (Hint: What can you conclude when $\epsilon = 1$?)

Solution: When $\epsilon = 1$, the definition of the limit tells is us there exists δ so that $|f(x) - L| < 1$ whenever $x \in N_0(a, \delta)$. Therefore we have $L - 1 < f(x) < L + 1$ for all $x \in N_0(a, \delta)$. Thus the set is bounded above by $L + 1$ and below by $L - 1$.

(c) State the contrapositive of the implication in the previous part.

Solution: If for every δ the set $\{f(x) \mid x \in N_0(a, \delta)\}$ is unbounded, then $f(x)$ diverges as x approaches a.

(d) Prove $\lim_{x \to 0} \dfrac{1}{x^2}$ diverges using the statement in the previous part.

Solution: Let $\delta > 0$, we claim the set $\{\frac{1}{x^2} \mid x \in N_0(0, \delta)\}$ is unbounded. We proceed by contradiction. Assume that the set is bounded above by some $r \in \mathbb{R}^+$. Therefore for all $x \in N_0(0, \delta)$ we have $x > \frac{1}{\sqrt{r}}$. However, for every $\delta > 0$ there exists $x \in \mathbb{R} \setminus \{0\}$ so that $x < \frac{1}{\sqrt{r}}$ and $x \in N_0(0, \delta)$, for example,

$$x = \frac{1}{2} \min\left\{\delta, \frac{1}{\sqrt{r}}\right\}.$$

Therefore the set is not bounded above by r.

5. CHALLENGE **Sequences and Functions Part III**

Let $f : E \to \mathbb{R}$ be a function, and let a be a limit point of E.

(a) Suppose that $\lim_{x \to a} f(x) = L$. Show that if (a_n) is a sequence such that $a_n \to a$ and $a_n \in E \setminus \{a\}$, then $\lim_{n \to \infty} f(a_n) = L$.

Solution: Let $\epsilon > 0$. By assumption,

$$\text{there exists } \delta > 0 \text{ such that for all } x \in E \setminus \{a\} \text{ if } |x - a| < \delta \text{ then } |f(x) - L| < \epsilon.$$

In turn, since $a_n \to a$,

$$\text{there exists } M \in \mathbb{N}^+ \text{ such that for all } n > M,\ |a_n - a| < \delta.$$

(This follows from the definition of convergent sequences, using δ as the "ϵ".) Combining the two, we have shown that

$$\text{there exists } M \in \mathbb{N}^+ \text{ such that for all } n > M,\ |f(a_n) - L| < \epsilon.$$

That is, we have shown that the sequence $(f(a_n))$ converges to L.

(b) Suppose now that for all sequences (a_n) converging to a and contained in $E \setminus \{a\}$, $\lim_{n \to \infty} f(a_n) = L$. Prove that $\lim_{x \to a} f(x) = L$.

Solution: We prove the contrapositive: If $\lim_{x \to a} f(x) \neq L$, then there exists a sequence (a_n) converging to a and contained in $E \setminus \{a\}$ such that $\lim_{n \to \infty} f(a_n) \neq L$. Suppose that $\lim_{x \to a} f(x) \neq L$. Then

$$\text{there exists } \epsilon > 0 \text{ such that for all } \delta > 0 \text{ there exists } x \in E \setminus \{a\} \text{ with } |x - a| < \delta \text{ but } |f(x) - L| \geq \epsilon.$$

Fix such an ϵ and for each $n \in \mathbb{N}^+$ choose $\delta_n = 1/n$ (so that $\delta_n > 0$ and $\delta_n \to 0$) and choose $a_n \in E \setminus \{a\}$ with $|a_n - a| < \delta$ and $|f(a_n) - L| \geq \epsilon$ (as above with $x = a_n$).

Claim 1: The sequence (a_n) converges to a.

Proof: Indeed, let $\epsilon' > 0$. By the Archimedean Principle, there exists $M \in \mathbb{N}^+$ such that $\delta_M = 1/M < \epsilon'$. From the way we chose a_n, it follows that for all $n > M$, $|a_n - a| < \delta_n < \delta_M < \epsilon'$.

Claim 2: The sequence $(f(a_n))$ does not converge to L.

Proof: Indeed, by construction of the sequence, we have $|f(a_n) - L| \geq \epsilon$ for all $n \in \mathbb{N}^+$. Thus, there is no $M \in \mathbb{N}^+$ such that for all $n > M$, $|f(a_n) - L| < \epsilon$.